

Zeigerrechnung Aufgaben

SIN1

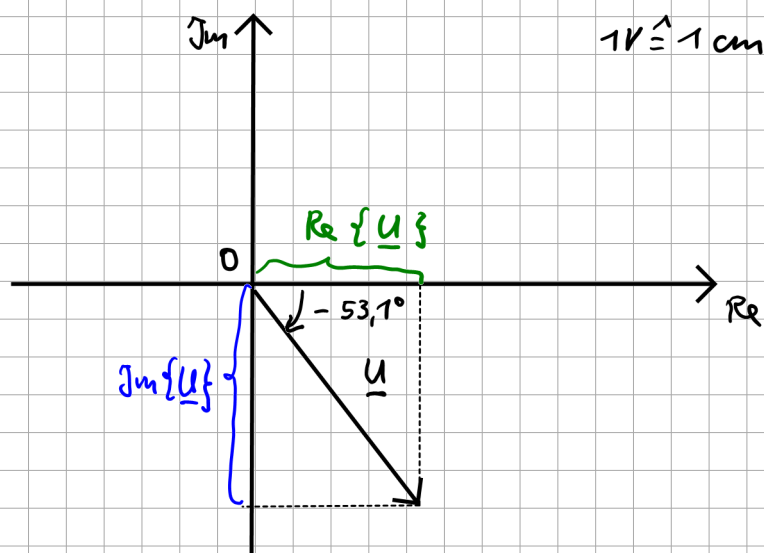
a) $u(t) = \hat{u} \cos(\omega t - 53,1^\circ) \quad \hat{u} = 5V$

$$\underline{U} = \frac{\hat{u}}{\sqrt{2}} \cdot e^{-j 53,1^\circ}$$

$$= \frac{5V}{\sqrt{2}} e^{-j 53,1^\circ}$$

$$= \underbrace{3,53V}_{\text{Betrag}} \cdot \underbrace{e^{-j 53,1^\circ}}_{\text{e-Fkt.}}$$

Exponentialform



Trigonometrische Form (sin, cos)

$$e^{jx} = \cos x + j \sin x$$

Euler-Gleichung

$$\underline{U} = 3,53V \cdot e^{-j 53,1^\circ}$$

$$= 3,53V \cdot [\cos(-53,1^\circ) + j \sin(-53,1^\circ)]$$

$$= 3,53V \cdot [0,6 - j 0,8]$$

$$= 2,12V - j 2,83V$$

Algebraische Form

SIN3

$$u_1(t) = 60V \cdot \cos(\omega t + \frac{3}{4}\pi)$$

$$f = 200\text{ Hz}$$

$$u_2(t) = 80V \cdot \cos(\omega t - \frac{1}{4}\pi)$$

a) Zeiger in Algebra-Form

$$\underline{u}_1 = \dots$$

$$\underline{u}_2 = \dots$$

$$\underline{u}_1 = \frac{60V}{\sqrt{2}} \cdot e^{+j\frac{3}{4}\pi}$$

$$= 42,43V \cdot e^{j135^\circ}$$

$$= 42,43V \cdot [\cos(135^\circ) + j\sin(135^\circ)]$$

$$= 42,43V \cdot [-0,707 + j0,707]$$

$$= -30V + j30V$$

$$2\pi \hat{=} 360^\circ$$

$$\frac{3}{4}\pi \hat{=} x$$

$$\Rightarrow x = \frac{\frac{3}{4}\pi \cdot 360^\circ}{2\pi} = 135^\circ$$

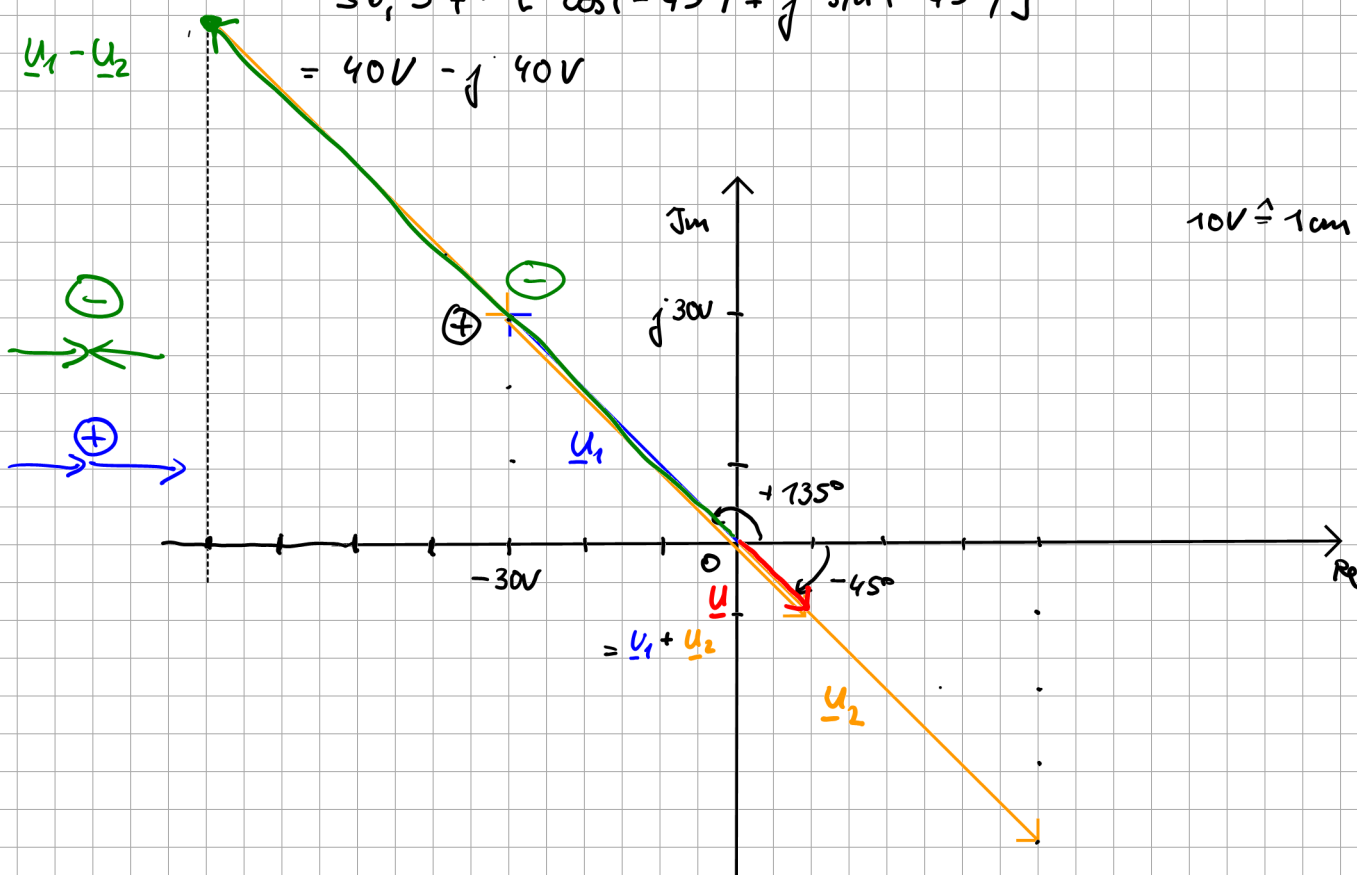
$$\underline{u}_2 = \frac{80V}{\sqrt{2}} \cdot e^{-j\frac{1}{4}\pi}$$

$$= 56,57V \cdot e^{-j45^\circ}$$

$$= 56,57 \cdot [\cos(-45^\circ) + j\sin(-45^\circ)]$$

$$= 40V - j40V$$

$$10V \hat{=} 1\text{ cm}$$



Ablezen: $\underline{U} = \underline{U}_1 + \underline{U}_2 = 10V - j10V$

$$= \sqrt{(10V)^2 + (10V)^2} \cdot e^{+j \arctan\left(\frac{-10}{10}\right)}$$

$$= \underbrace{\sqrt{2} \cdot 10V}_{=U} \cdot e^{-j45^\circ}$$

$$u(t) = \hat{u} \cdot \cos(\omega t + \varphi)$$

$$= \sqrt{2} \cdot (\sqrt{2} \cdot 10V) \cdot \cos(\omega t - 45^\circ)$$

$$\underline{U} = \underline{U}_1 - \underline{U}_2$$

$$= (-30V + j30V) - (40V - j40V)$$

$$= \ominus 70V + j70V$$

$$= \sqrt{(-70V)^2 + (70V)^2} \cdot e^{j \arctan\left(\frac{70V}{-70V}\right) \pm 180^\circ}$$

falls $R_e < 0$!

$$= \sqrt{2} \cdot 70V \cdot e^{-j45^\circ}$$

↗

$$\underline{z} = a + jb$$

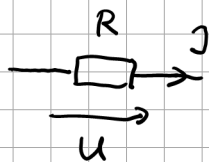
$$\underline{z} = \sqrt{a^2 + b^2} \cdot e^{j \arctan\left(\frac{b}{a}\right) \pm 180^\circ}$$



$$\varphi = \arctan\left(\frac{b}{a}\right)$$

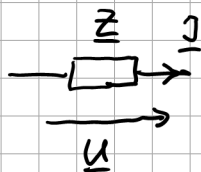
falls $a < 0$

SIN 5



$$U = R \cdot I$$

komplexer Widerstand „Impedanz“ $\underline{Z}$



$$\underline{U} = \underline{Z} \cdot \underline{I}$$

Bsp: $\underline{Z} = 2k\Omega + j1k\Omega$

$\underline{I} = 4mA + j3mA$

$\underline{U} = \dots$

(Algebra-Form, Exponential-Form)

$\underline{U} = \underline{Z} \cdot \underline{I}$

$= (2k\Omega + j1k\Omega) (4mA + j3mA)$ Algebra

$= (2 + j1) \cdot (4 + j3) k\Omega \cdot mA$

$= (2 + j1) \cdot (4 + j3) V$

$= (8 + j4 + j6 + j^2 3) V$

$= (8 - 3 + j10) V$

$= 5V + j10V$

$\Omega \cdot A = \frac{V}{A} \cdot A = V$

$\underline{U} = \sqrt{(5V)^2 + (10V)^2} \cdot e^{j \arctan(\frac{10V}{5V})}$

$= 11,18V \cdot e^{j63,4^\circ}$

c)

$\underline{Z} = 2k\Omega + j1k\Omega$

(Algebra)

↓
Exp. Form

$= \sqrt{(2k\Omega)^2 + (1k\Omega)^2} \cdot e^{j \arctan(\frac{1k\Omega}{2k\Omega})}$

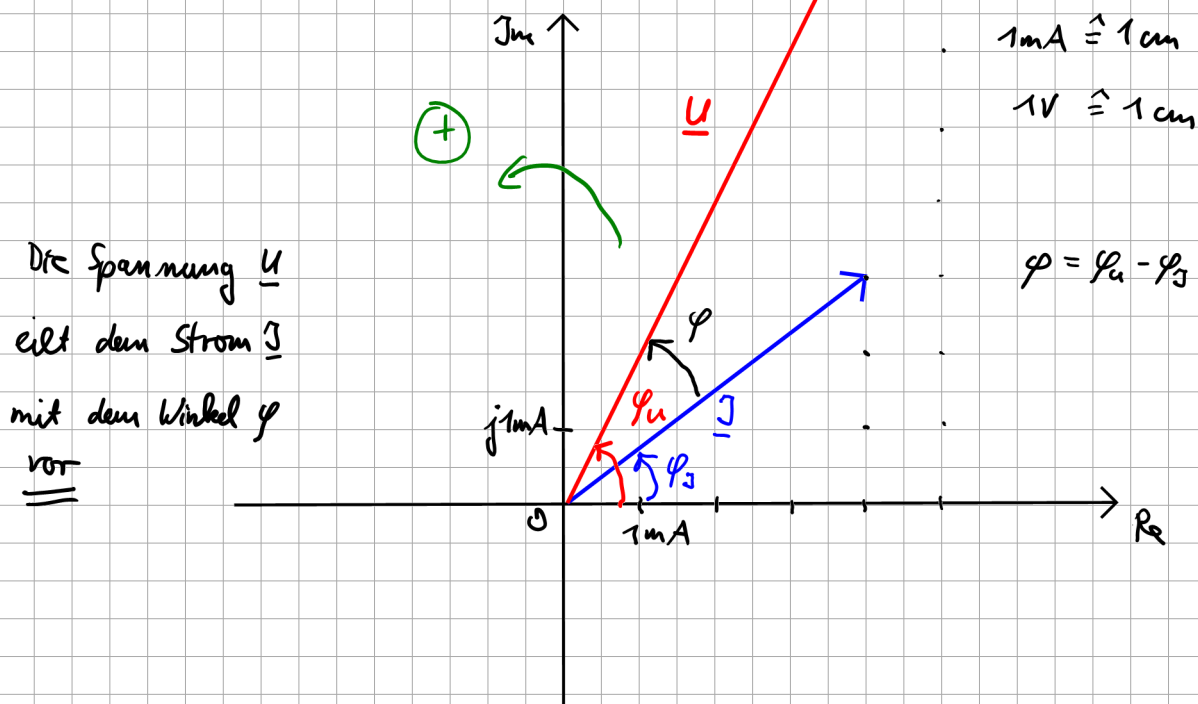
$= \sqrt{5} k\Omega \cdot e^{j26,6^\circ}$

$\underline{I} = 4mA + j3mA$

$= \sqrt{4^2 + 3^2} mA \cdot e^{j \arctan(\frac{3}{4})}$

$\underline{I} = 5mA \cdot e^{j36,9^\circ}$

$$\begin{aligned}
 \underline{U} &= \underline{Z} \cdot \underline{I} \\
 &= \sqrt{5} \, k\Omega \cdot e^{j26,6^\circ} \cdot 5 \, mA \cdot e^{j36,9^\circ} \\
 &= 11,18 \, V \cdot e^{j(26,6^\circ + 36,9^\circ)} \\
 &= 11,18 \, V \cdot e^{j63,5^\circ}
 \end{aligned}$$



$$\underline{U} = 11,18 \, V \cdot e^{j63,5^\circ}$$

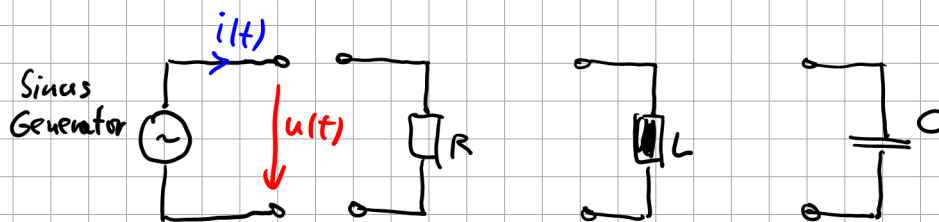
$$\begin{aligned}
 u(t) &= \sqrt{2} \cdot 11,18 \, V \cdot \cos(\omega t + 63,5^\circ) \\
 &= 15,81 \, V \cdot \cos(\omega t + 63,5^\circ)
 \end{aligned}$$

$$\underline{I} = 5 \, mA \cdot e^{j36,9^\circ}$$

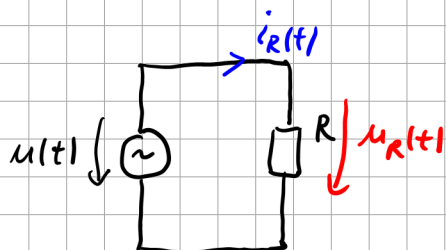
$$\begin{aligned}
 i(t) &= \sqrt{2} \cdot 5 \, mA \cdot \cos(\omega t + 36,9^\circ) \\
 &= 7,07 \, mA \cdot \cos(\omega t + 36,9^\circ)
 \end{aligned}$$

3 R, L, C bei Sinusanregung

Gedanke:



Ohmscher Widerstand R an Sinusquelle:



$$u_R(t) = R \cdot i_R(t)$$

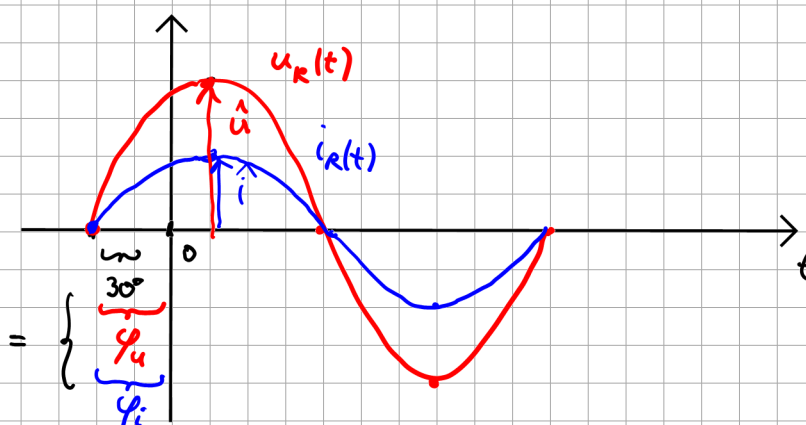
Ohmsches Gesetz
(gilt immer)

Sei: $u(t) = \hat{u} \cdot \cos(\omega t + \varphi)$

$$u(t) = u_R(t) = R \cdot i_R(t)$$

$$\Rightarrow i_R(t) = \frac{u_R(t)}{R} = \frac{\hat{u}}{R} \cdot \cos(\omega t + \varphi) = \hat{i} \cdot \cos(\omega t + \varphi)$$

φ_i



$$\varphi_u = \varphi_i$$

